Quantum Measurements as Minkowski Microstates

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Abstract

We show how quantum state reduction naturally plays the role of a microstate in thermodynamic approaches to gravity and develop an information-theoretic language to explore the connection with a concrete construction. Eliminating temperature between Landauer's principle and the Unruh effect yields a purely kinematic acceleration-dependent bound on the energy cost of information transfer. Motivated by this, we construct a helicity-anticorrelated Bell state in flat Minkowski space, emitted from a common causal past, that drives the Unruh response in both Rindler wedges. We briefly discuss how proper time can be measured as a property of matter, and then connect the matter action, event by event, to a Gibbons–Hawking–York boundary term and a horizon area increment. Using a pp-wave cut-and-paste surgery, we show how space is dynamically advanced along the horizons. This newly generated geometry carries a concrete label tied to the entangled helicity, which is subsequently observed across the causal horizons, allowing for the measurement of space in geometric and informational terms simultaneously.

1 Introduction

Jacobson derives the Einstein equations from entropy and temperature on local Rindler horizons [1]; Verlinde derives gravitational forces from entropy gradients on holographic screens [2]. Both frameworks operate at the level of bulk thermodynamics and leave open a prior question: *what physical process underlies each individual entropy increment?* We will show how the answer can be quantum measurement by developing an information-theoretic language that connects "thrust" microstates, that can drive the Unruh effect [3], to current literature.

Before we dig in though, we first briefly outline our high level argument about how quantum measurement naturally connects to gravity. The general setup is pictured in the commutative[1] diagram in Figure 1, which we will walk through from the top right clockwise around the bottom to the top left. Quantum state reduction always happens due to an observation. Any observation is mediated by a physical interaction: information transfer necessarily involves an exchange of energy and momentum between the observed system and the detector, a discrete impulse. And then the equivalence principle implies that this impulse is locally indistinguishable from the effect of a gravitational field.

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[1]This refactors the Diósi–Penrose postulate in the categorical sense: the dashed arrow is not assumed but derived as the composite of the three solid arrows. Diósi–Penrose postulates a direct link from gravitational effects to state reduction; the present diagram factors that link through acceleration and observation, grounding each step independently.

$$
\begin{tikzcd}
\text{Local Gravitational Effects} \arrow[r, dashed, "\text{Diósi–Penrose}"] & \text{Quantum State Reduction} \\
\text{Acceleration} \arrow[u, "\substack{\text{local equivalence}\\ \S5}"'] \arrow[r, "\substack{\text{energy–momentum}\\ \text{transfer}\\ \S2}"'] \arrow[ur, phantom, "\circlearrowright\ \S4"] & \text{Observation} \arrow[u, "\substack{\text{Bayesian}\\ \text{update}\\ \S3,\S6}"']
\end{tikzcd}
$$

Figure 1: Solid arrows denote subjective grounding relations for a given observer: $A \to B$ means every instance of B is physically realized by an instance of A. This paper explores the solid arrows. The dashed arrow is the Diósi–Penrose postulate.

1.1 Roadmap

The paper develops each solid arrow of Figure 1 in turn, with Section 4 providing the information-theoretic framework that unifies them into a single physical unit, a communication channel between massive observers mediated by a massless quantum.

The **energy–momentum transfer** arrow is developed in Section 2: substituting the Unruh temperature into Landauer's principle eliminates temperature entirely, yielding a purely kinematic bound on the energy cost of information acquisition under acceleration. The thrust quantum, in our case a photon carrying the impulse, is the physical carrier of this bound.

The **Bayesian update** arrow is developed in Sections 3 and 6: the POVM-to-PVM transition selects a single non-thermal microstate from the Minkowski vacuum, the helicity outcome is recorded as a spatial bit in matter memory, and the anti-correlation between right and left Rindler wedges is enforced by their common causal ancestry in the past wedge.

The **local equivalence** arrow is developed in Section 5: the absorbed photon sources an Aichelburg–Sexl pp-wave at the Rindler horizon, the Raychaudhuri equation gives a discrete area increment $\Delta A = 8\pi G\hbar\omega$, and the event-by-event equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ is energy conservation written simultaneously in matter and geometric language, with the Planck area ℓ_P^2 as the conversion factor.

Section 4 is the conceptual core. It explains why these three arrows are aspects of one physical unit: a communication that transfers action and information between massive observers. Several of the thermodynamic identities recovered along the way, the area law, the Bekenstein bound, the Unruh spectrum, are well established; what is new is locating them microstate by microstate in a single physical event, rather than recovering them only as a bulk average.

2 Observation and Acceleration

The elimination of temperature in this section conceptually and physically leads the way to Section 3, where we construct concrete non-thermal microstates that drive the Unruh response, and the later sections that connect our microstate to the broader literature. It allows us to work in the non-thermal global Minkowski description, where thermality is a derived consequence of restricting to one wedge [4, 5] rather than a primitive.

Section 2.1 derives the Unruh–Landauer bound as an exact kinematic result: substituting the Unruh temperature into Landauer's principle eliminates temperature entirely, leaving a purely kinematic inequality depending only on acceleration, which the construction of Sections 3–5 inherits directly. Section 2.2 identifies the physical carrier of this bound as a thrust quantum and connects the resulting momentum transfer to Verlinde's entropic gravity program. Verlinde-style heuristics recovering standard black hole thermodynamics from this bound are collected in Appendix B.

2.1 The Unruh–Landauer Relation

Landauer's principle [6] establishes $k_B T$ as the natural energy scale of one distinguishable degree of freedom in a thermal environment at temperature T: any physical carrier representing ΔH nats must have at least this energy to be distinguishable from thermal noise,

$$\Delta E \geq k_B T \, \Delta H. \tag{1}$$

For an observer undergoing proper acceleration a, the Unruh effect associates a temperature

$$T = \frac{\hbar a}{2\pi k_B c} \tag{2}$$

with the local vacuum. Substituting into Landauer's bound eliminates $k_B T$, yielding

$$\boxed{\Delta E \geq \frac{\hbar a}{2\pi c} \, \Delta H.} \tag{3}$$

Temperature has dropped out entirely: equation (3) is a temperature-independent lower bound on the energy of the physical carrier required to register one nat of information under proper acceleration a; acceleration sets the minimum energy scale of the carrier. The full thermodynamic accounting, where the Landauer erasure cost appears, is deferred to future work.

The connection between Landauer's principle and black hole thermodynamics has been noted previously. Cortês and Liddle [7] showed that Hawking evaporation saturates the Landauer bound at the Hawking temperature, identifying the two results as expressions of the same process at the level of bulk evaporation. Related bounds on the energy cost of information acquisition in curved spacetime have been derived using the second law of thermodynamics applied to relativistic communication channels [8]. The present bound (3) differs in three respects: temperature is eliminated entirely by substituting the Unruh relation, yielding a purely kinematic bound depending only on acceleration; the bound is applied to individual carrier quanta rather than to bulk evaporation; and the microstates are constructed explicitly in Sections 3–5 rather than recovered as bulk averages.

This Unruh–Landauer bound is saturated for a photon at one bit when the wavelength equals the horizon distance up to a factor of $4\pi^2/\ln 2$:

$$\lambda_{\text{sat}} = \frac{4\pi^2}{\ln 2} \cdot \frac{c^2}{a} \sim d_{\mathcal{H}}, \tag{4}$$

where $d_{\mathcal{H}} = c^2/a$ is the proper distance to the Rindler horizon. The saturating photon is the fundamental mode of the causal diamond defined by the observer and the horizon, with wavelength commensurate with the horizon distance $d_{\mathcal{H}}$, too short to be super-horizon and too long to deposit more than one bit. This identifies the horizon distance as the natural length scale of a single-bit measurement event, and the Rindler horizon as the surface at which the Landauer bound is geometrically realized.

For realistic accelerations the saturation wavelength is enormous, at Earth surface gravity $a = 9.8\ \mathrm{m\,s^{-2}}$, $\lambda_{\text{sat}} \approx 55$ light years, so the single-bit bound is practically always satisfied for terrestrial observers, and is only approached near astrophysical or Planck-scale accelerations.

2.2 Momentum Transfer and Acceleration

The energy transfer ΔE implied by the Unruh–Landauer relation is carried by a physical quantum, matter or radiation, mediating the measurement interaction as a *thrust quantum.* For a relativistic carrier, the associated momentum transfer is

$$\Delta p = \frac{\Delta E}{c}. \tag{5}$$

This is closely related to Verlinde's entropic gravity [2]: Verlinde identifies holographic thrust on information screens with the origin of inertia itself, and Newton's laws are a common ingredient in both constructions. The present approach differs in grounding each entropy increment in a discrete measurement event, the measurement impulse playing the role of Verlinde's holographic thrust, rather than in a coarse-grained thermodynamic gradient.

The Unruh–Landauer bound is saturated not at the level of an individual photon but at the level of the source rate[2]. The acceleration a does not fix the frequency ω of each selected quantum; rather, it determines the rate at which the source selects entangled pairs in order to implement that acceleration, with the total power $\dot{N}\hbar\omega$ matching the Unruh thermal flux. Each individual photon carries exactly one quantum of action $\hbar$ and one binary helicity outcome, one bit in the sense of one binary alternative carrying $\ln 2$ nats, and these are paired by the construction independently of a. The Landauer bound then operates as a consistency condition on the ensemble: the minimum energy cost per bit acquired, summed over the rate of acquisition, equals the power required to sustain the acceleration.

Newton's second law gives the interaction rate directly for a detector of mass m at rest:

$$a = \frac{\dot{N}(a)\,\hbar\omega}{mc}, \qquad \dot{N}(a) = \frac{mca}{\hbar\omega}. \tag{6}$$

Two consequences of this rate and the bound (3) are developed later: the ensemble of individual non-thermal microstates selected at rate $\dot{N}(a)$ is shown in Section 4.3 to saturate the channel capacity exactly, providing a dynamical account of how thermality emerges from coarse-graining over individual measurement events; and the bound (3) is shown in Appendix B.3 to coincide with the Bekenstein universal entropy bound at the horizon condition $a = c^2/R$, identifying the two as the same inequality in acceleration-frame and position-space descriptions respectively.

3 Bilocal Microstate Construction

From now on we use natural units $\hbar = c = 1$ unless otherwise specified; factors of $\hbar$ and c are restored where they aid physical interpretation.

The arguments of Section 2 motivate a relationship between information acquisition, acceleration, and gravitational effects, but leave open the question of what a single microstate looks like concretely. In standard treatments, the Unruh effect is derived by coupling a detector to a pre-existing acceleration and reading off a thermal response from the vacuum. In [3] this logic was inverted for a scalar field: a source placed in the past injects non-thermal entangled excitations into the field, and the observer's accelerated response was shown to be driveable by these source-induced excitations. We extend that construction here to the spin-1 electromagnetic field, upgrading from scalar to helicity-carrying Bell states and making the single-bit structure of each measurement event explicit.

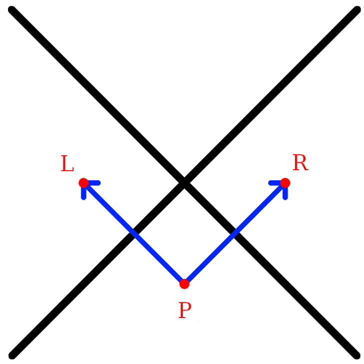

Figure 2: A source in the past, $\mathcal{P}$, emits an entangled photon pair that accelerates $\mathcal{R}$ in the right Rindler wedge and $\mathcal{L}$ in the left Rindler wedge. The two null directions of propagation define the null coordinates $u = t-z+1$ and $v = t+z+1$, with the right-moving photon propagating along $u = 0$ and the left-moving photon along $v = 0$; these are the z^- and z^+ axes of the double-null coordinate system of Section 3.4.

3.1 A Source and Two Observers

Three massive bodies are placed at spacetime positions $(t, z) = (0, 1)$, $(0, -1)$, and $(-1, 0)$, named $\mathcal{R}$ – *Right*, $\mathcal{L}$ – *Left*, and $\mathcal{P}$ – *Past* respectively. $\mathcal{P}$ serves as the common causal ancestor of the accelerations of $\mathcal{R}$ and $\mathcal{L}$ (see Figure 2). $\mathcal{P}$ emits an entangled pair of spin-1 massless photons with momenta $\pm k$ and Rindler frequency $\omega = \omega_k$. The source exhibits no net back-reaction in either linear or angular momentum[3]. The two photons propagate into the left and right Rindler wedges, where $\mathcal{L}$ and $\mathcal{R}$ detect them respectively via absorption. The momentum impulse accelerates each detector in its respective Rindler wedge; each observer thereby registers a helicity eigenvalue and acquires one classical bit of information. The emission event at $\mathcal{P}$ naturally defines null coordinates $u = t - z + 1$ and $v = t + z + 1$, centered so that $\mathcal{P}$ lies at $u = v = 0$, the right-moving photon propagates along $u = 0$, and the left-moving photon propagates along $v = 0$. The three-party structure $\mathcal{P}$–$\mathcal{R}$–$\mathcal{L}$ realises a form of tripartite entanglement[4] with a natural interior reconstruction interpretation [16, 17].

3.2 PVM Localization and the Elimination of Coarse-Graining

The Unruh mode plays a double role in this construction. As a QFT object it is a coefficient in a mode expansion, a solution to the wave equation analytically continued from the Rindler wedge into all of Minkowski space. But once $\mathcal{P}$ is specified as a real physical source via the squeezing interaction (equation (14) upcoming), the mode function acquires a second interpretation: it is a probability amplitude in the Born rule sense,

[2] Here we discuss a constant rate with a fixed frequency, but many situations are possible.

[3] The momenta are perfectly balanced by construction: the two photons carry spatial momenta $+k\hat{z}$ and $-k\hat{z}$ respectively, so the total linear momentum vanishes, $k + (-k) = 0$. The angular momentum vanishes term by term: in each helicity assignment $(\lambda_R, \lambda_L) = (+1, -1)$ and $(-1, +1)$, the total helicity is $\lambda_R + \lambda_L = (+1) + (-1) = 0$ and $(-1) + (+1) = 0$ respectively, and the symmetric Bell state superposition (20) preserves this since both terms contribute zero independently

[4] Neither $\mathcal{R}$ nor $\mathcal{L}$ alone can distinguish a correlated Bell-state emission at $\mathcal{P}$ from an uncorrelated thermal source: each observer's reduced density matrix is thermal, exactly as in the standard Unruh effect. Together, their two anticorrelated bits reveal the correlation structure of the emission event, identifying it as a specific Bell state at a specific frequency, and providing a microstate-level analogue of black hole interior reconstruction, where the interior is recovered as classical correlations in the joint record of complementary exterior observers.

encoding where $\mathcal{R}$ is likely to detect the emitted quantum. This is the inversion of the standard Unruh–DeWitt logic. Rather than coupling a detector to a pre-existing acceleration and reading off a thermal distribution by averaging over all Rindler frequencies, we front-load the logic at $\mathcal{P}$: the source fixes the frequency ω_k and the mode function then specifies the probability amplitude for detection.

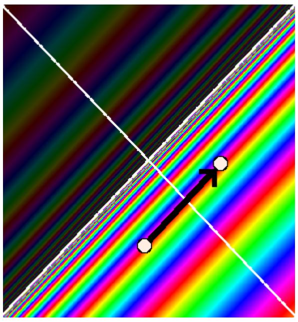

Figure 3: The points $\mathcal{P}$ and $\mathcal{R}$ are marked with white dots, connected by the null geodesic (black arrow). The rainbow coloring shows the Unruh mode phase structure across Minkowski space. A PVM projection onto this null line selects a single frequency from the thermal superposition, replacing the full delocalized mode with the localized non-thermal microstate of equation (20).

Longitudinal localization. The primary localization is in the longitudinal (t, z) plane, which is the setting of Figure 3 and of the parabolic cylinder function analysis of [3]. In that work, a Minkowski mode φ_q^* multiplied pointwise by a Gaussian shaping function $N_{\mu,\sigma}$ of width σ in (t, z), then paired via the Klein–Gordon inner product with a Rindler mode r_k, yields an overlap expressible in terms of parabolic cylinder functions $D_\nu(-\zeta)$. The Gaussian could equivalently have been applied to the Rindler side; what matters is that σ controls the degree of longitudinal localization of the source-detector pair. The Stokes phenomenon in the asymptotics of $D_\nu(-\zeta)$ as σ varies drives a transition between two physically distinct regimes: $\sigma \to \infty$ gives the thermal regime where the $e^{-\frac{1}{4}\zeta^2}$ branch dominates, and $\sigma \to 0$ gives the localized non-thermal regime where the $e^{+\frac{1}{4}\zeta^2}$ branch dominates, with the Stokes line at $\arg \zeta = \pi/4$ marking the boundary between them.

In the present paper the same Gaussian shaping function is reread as a Kraus operator acting on the quantum state.

$$K_\sigma = \int e^{-u^2/2\sigma^2} \, |u\rangle\langle u| \, du, \tag{7}$$

where $u = t - z + 1$ is the null coordinate[5] defined in Section 5. The operator K_σ together with its complement

$$K_\sigma^\perp = \int \sqrt{1 - e^{-u^2/\sigma^2}} \, |u\rangle\langle u| \, du, \tag{8}$$

satisfying $K_\sigma^\dagger K_\sigma + (K_\sigma^\perp)^\dagger K_\sigma^\perp = \mathbf{1}$ by construction, defines a one-parameter POVM family; the Stokes transition between its two asymptotic regimes is illustrated in Figure 4.

The Stokes transition of [3] is therefore a field-theoretic regularization result and the analytic signature of the POVM-to-PVM transition: the same mathematical object is being read in two complementary languages, field theory and measurement theory, with the Stokes line marking the boundary between them. Specifically, the result imported

[5] We use ζ for the Stokes variable of [3] to avoid confusion with the longitudinal spatial coordinate z.

from [3] is that the overlap between a Gaussian-shaped Minkowski source of width σ and a Rindler mode at frequency ω is expressible in terms of parabolic cylinder functions $D_\nu(-\zeta)$, whose asymptotic behaviour undergoes a Stokes transition at $\arg\zeta = \pi/4$. In the present paper this result is reread as the operator statement that the Kraus family (7) interpolates between a POVM ($\sigma \to \infty$) and a PVM ($\sigma \to 0$), with the Stokes line marking the boundary between the two regimes.

The helicity label λ enters the longitudinal mode function as a spectator index and does not affect the u-dependence of the Kraus operator (7); the Stokes transition analysis of [3], which is carried out in 1+1 dimensions, therefore applies independently to each helicity sector without modification. The extension to 3+1 dimensions introduces the transverse coordinates (x, y) as an independent sector: because the wave equation in double-null coordinates separates, the longitudinal Stokes analysis is unaffected by the transverse directions, which are handled separately by the transverse Kraus family equation (9).

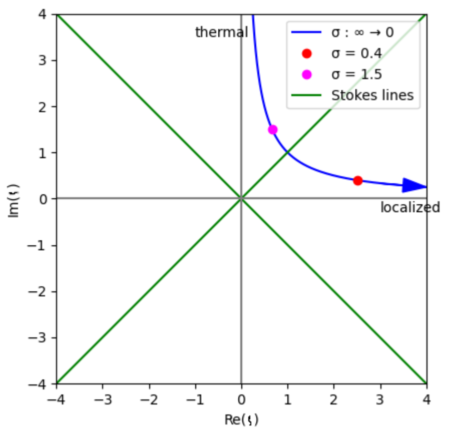

Figure 4: Trajectory of $\zeta = iq\sigma + \mu/\sigma$ in the complex plane as σ interpolates between the two limits of the Kraus family (7). The thermal limit $\sigma \to \infty$ corresponds to ζ on the positive imaginary axis; the localized PVM limit $\sigma \to 0$ corresponds to ζ on the positive real axis. The green Stokes lines at $\arg\zeta = \pi/4$ mark the boundary between the two asymptotic regimes of the parabolic cylinder function $D_\nu(-\zeta)$, and therefore the boundary between the POVM and PVM descriptions of the longitudinal localization. Figure from [3].

This distinction is operationally sharp [18]: *mode shaping* is a preparation that modifies the physical field; *PVM projection* is a measurement that conditions the observer's description on causal data without touching the ontic state. The Gaussian width σ of the Kraus family (7) admits both readings simultaneously, as a source localization parameter in the preparation sense of [3], and as an epistemic resolution parameter in the measurement sense here, with the Stokes line marking the boundary at which the two readings give qualitatively different physics. Specifying $\mathcal{R} = (0, 1)$ and $\mathcal{P} = (-1, 0)$ is therefore the epistemic $\sigma \to 0$ limit: the Kraus operator sharpens to a PVM projector onto the unique null geodesic connecting them, selecting a definite frequency ω_k and resolving both coarse-grainings responsible for thermality. The direction of explanation is thereby reversed relative to standard holographic arguments: the emission event at $\mathcal{P}$ is primary, and the geometry, the AS pp-wave, the horizon area increment, the BMS supertranslation, is the ontic output of the subsequent measurement events, while the assignment of that output to a particular observer's microstate record remains epistemic.

Transverse localization. The transverse (x, y) directions require a separate and independent localization. A Rindler plane wave at frequency ω is a coherent superposition over all possible source-detector pairs $(\mathcal{P}', \mathcal{R}')$ connected by null geodesics at that frequency, uniform in the transverse plane. The longitudinal and transverse localizations decouple because the wave equation in double-null coordinates separates: the u-dependence and the (x, y)-dependence of the mode function enter independently, so the two Kraus projections can be applied in either order without interference. Specifying $\mathcal{P} = (-1, 0)$ and $\mathcal{R} = (0, 1)$ performs a second PVM projection, now in the transverse directions, selecting the unique null geodesic and resolving the remaining coarse-graining over transverse position. The transverse shaping is controlled by an independent Kraus family

$$K_{\sigma_\perp} = \int e^{-r^2/2\sigma_\perp^2} \, |\mathbf{x}_\perp\rangle\langle\mathbf{x}_\perp| \, d^2x_\perp, \tag{9}$$

the direct transverse analogue of (7). The total horizon area increment $\Delta A = 8\pi G\hbar\omega$ is independent of $\sigma_\perp$ by energy conservation, as the Raychaudhuri calculation of Section 5 shows: only the total null energy $\int T_{uu} \, du \, d^2x_\perp = \hbar\omega$ enters, not the transverse distribution. We work in the localized limit $\sigma_\perp \to 0$, which recovers the Aichelburg–Sexl pp-wave developed in Section 5. The BMS mode content excited by different transverse profiles, and the physical selection principle for $\sigma_\perp$, are left to future work.

The area increment $\Delta A = 8\pi G\hbar\omega$ and the event-by-event equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ in Section 5, which serve as the geometric consistency checks of the framework, are independent of $\sigma_\perp$, fixed by energy conservation alone.

The microstate. The result of both projections is an individually non-thermal microstate carrying a definite Rindler frequency ω and a definite longitudinal position on the null geodesic from $\mathcal{P}$ to $\mathcal{R}$. Thermality is not a property of any single microstate but of the ensemble; it emerges when one models the detector response within a wedge as a sum over all Rindler modes, as in the standard Unruh–DeWitt treatment [11, 19], and is a consequence of that mode-averaging rather than a premise of the construction. The individual non-thermal microstate is what the PVM projection onto the $\mathcal{P}$–$\mathcal{R}$ null geodesic selects; the thermal ensemble is what remains when neither coarse-graining is resolved and $\mathcal{L}$ is traced out.

Throughout this paper the PVM limit is understood as $\sigma \to 0^+$: the Kraus operator (7) is not normalizable at exactly $\sigma = 0$, and the AS shift $\Delta v \propto -\ln r^2$ diverges on axis, both regularized by the photon wavelength $\sigma \sim c/\omega$. The parameter σ is simultaneously the longitudinal Kraus width, the transverse profile width, and the AS regularization scale.

3.3 Helicity Structure and PCT Symmetry

We now upgrade this construction from a spin-0 scalar field in [3] to the spin-1 electromagnetic field with a superposition of helicity pairs in a Bell state, introducing one unknown binary degree of freedom, the helicity outcome, undetermined until measured, which is the single bit responsible for the remaining thermality in the ensemble. The photon carries one quantum of action $\hbar$ and energy $E = \hbar\omega$, identifying the thrust quantum of Section 2 with a single photon whose polarization outcome, upon detection, is the classical bit acquired by each observer. The Bell state is the natural completion of the PCT symmetry between wedges, which by the Bisognano–Wichmann theorem [5] is implemented by the modular conjugation operator J mapping the right wedge algebra to the left wedge algebra, and as we discuss in Section 5, the photons imprint a definite geometric signature onto the horizon.

The anticorrelated helicity assignment traces to the PCT symmetry implemented by the Bisognano–Wichmann modular conjugation J, which maps the right-wedge algebra to the left-wedge algebra and simultaneously flips $\lambda \to -\lambda$; the Bell state (20) is the natural state in this momentum sector invariant under that operation. The common algebraic origin of this anticorrelation in the Lorentz algebra decomposition $\mathfrak{so}(3,1) \cong \mathfrak{su}(2) \times \mathfrak{su}(2)$, developed in Appendix A.

3.4 Double-Null Gauge

The emission event naturally defines two null directions: the right-moving photon has wave-vector $k_R^\mu = (\omega, 0, 0, k)$ and the left-moving photon has wave-vector $k_L^\mu = (\omega, 0, 0, -k)$, with equal and opposite spatial momenta. These two null directions are precisely the z^+ and z^- axes of double-null coordinates

$$z^\pm = \frac{1}{\sqrt{2}}(t \pm z) = \frac{1}{\sqrt{2}}(u - 1 \pm (v - 1)), \tag{10}$$

with transverse coordinates $\mathbf{z}_\perp = (x, y)$. Note that $z^\pm$ are centered on the Minkowski origin $(t, z) = (0, 0)$ rather than on $\mathcal{P}$; the coordinates u, v defined in Section 3 are the $\mathcal{P}$-centered versions and are used throughout the remainder of the paper. Working in this frame we impose the symmetric double-null gauge condition

$$A_+(z) = 0, \qquad A_-(z) = 0, \tag{11}$$

so that the gauge field is supported entirely on the transverse components $\mathbf{A}_\perp = (A^x, A^y)$. This gauge respects the PCT symmetry that swaps the two wedges ($z^+ \leftrightarrow z^-$) simultaneously with ($R \leftrightarrow L$), and therefore treats both observers symmetrically by construction. The residual gauge freedom is fixed by requiring $\partial_i A^i = 0$ in the transverse plane, leaving precisely the two physical helicity degrees of freedom $\epsilon^i_{(\lambda)}$ with $\lambda = \pm 1$.

The lack of manifest four-dimensional covariance in equation (11) is a consequence of aligning the gauge with the pair of physical null directions selected by the source, rather than an arbitrary choice. The full Lorentz group is broken to the little group preserving the pair $\{k_R^\mu, k_L^\mu\}$, which is the two-dimensional Euclidean group $ISO(2)$ acting on the transverse plane. This is precisely the little group of massless particles, and the two helicity eigenstates $\lambda = \pm 1$ are its irreducible representations. All physical results are independent of the gauge choice: the helicity eigenvalues $\lambda = \pm 1$ are representations of the massless little group $ISO(2)$ and are therefore Poincaré-covariant objects independent of how the gauge is fixed.

3.5 The Bilocal Source

In double-null gauge the electromagnetic field is characterised by its transverse components alone. The bilocal driving source is symmetrized over both helicity assignments,

$$J^{ij}(x, y) = g \left(\epsilon^{i*}_{(-1)} \, \epsilon^{j}_{(+1)} + \epsilon^{i*}_{(+1)} \, \epsilon^{j}_{(-1)} \right) f_L(x) \, f_R(y), \tag{12}$$

where f_L and f_R are normalised mode profiles supported on the left and right wedges respectively, and $\epsilon^i_{(\lambda)}$ are the circular polarization vectors

$$\epsilon^i_{(+1)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \qquad \epsilon^i_{(-1)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}. \tag{13}$$

The two terms in (12) are related by the PCT transformation in the sense of Bisognano–Wichmann [5], which maps the right wedge algebra to the left wedge algebra and simultaneously flips $\lambda \to -\lambda$.

More precisely, the modular conjugation J of Bisognano–Wichmann maps a right-wedge creation operator $a^{R\dagger}_{k,(\lambda)}$ to a left-wedge creation operator $a^{L\dagger}_{-k,(-\lambda)}$, implementing PCT between wedges. The Bell state (20) is a natural two-photon state at fixed frequency

ω and fixed momentum pair $(k, -k)$: within this momentum sector, the anticorrelated helicity assignment and the symmetric superposition respect the J-symmetry between wedges, since J flips both helicity and wedge simultaneously. The bilocal source (12) is therefore a natural source, consistent with the algebraic symmetry of the Rindler vacuum, and the simplest one we could construct with the required properties.

The symmetrization is consistent with PCT symmetry and angular momentum conservation at the emission vertex. Other source geometries are possible; this one is chosen for its simplicity and the transparency of its connection to the algebraic structure of the Rindler vacuum. A scalar source carries zero angular momentum, so the emitted pair must satisfy $\lambda_R + \lambda_L = 0$, which forces the two helicities to be anticorrelated: the only possibilities are $(\lambda_R, \lambda_L) = (+1, -1)$ or $(-1, +1)$. The Bisognano–Wichmann modular conjugation J maps the first assignment to the second and vice versa, and the symmetric superposition of the two is the simplest state respecting this exchange symmetry. The total angular momentum vanishes in both terms independently.

The bilocal source selects a specific microstate from the Minkowski vacuum by a PVM projection onto the $n = 1$ sector at fixed ω and fixed helicity pair $(\lambda, -\lambda)$, in exact analogy with the scalar construction of [3]. No perturbative expansion in a coupling constant is required: the helicity-anticorrelated photon pair is already present in the vacuum decomposition (18), and the bilocal source selects it via the PVM projection of Section 3.2. The operator content of this selection is

$$a^{L\dagger}_{-k,(-1)}\, a^{R\dagger}_{k,(+1)} \;+\; a^{L\dagger}_{-k,(+1)}\, a^{R\dagger}_{k,(-1)} \;+\; \text{h.c.}, \tag{14}$$

where the two creation operator pairs correspond to the two PCT-related helicity assignments, each carrying total angular momentum zero, consistent with emission from a scalar source at $\mathcal{P}$. This is the vector-field analogue of the scalar squeezing interaction [20], with the PCT symmetry between wedges made explicit; the connection to the quantum optics squeezing literature is structural rather than perturbative.

Each photon carries energy $E = \hbar\omega$ and one quantum of action $\hbar$, identifying the thrust quantum of Section 2 with a concrete physical object: the minimum actionable event is precisely a single photon. Its polarization, undetermined until measured, is the quantum bit whose outcome constitutes the classical record acquired by each observer upon detection.

The Unruh–Landauer bound and the interaction rate of Section 2.2 apply directly to the bilocal construction, with $m = m_{\mathcal{R}}$ and the source selecting helicity-anticorrelated photon pairs at rate $\dot{N}(a)$. Each photon carries exactly one quantum of action $\hbar$ and one binary helicity outcome; the anticorrelation between wedges is enforced by the Bell state structure (20). The saturation condition of Section 4.3 then follows directly.

3.6 Polarization Bogoliubov Transformation

We now show that the polarization labels survive the Rindler–Unruh mode decomposition, so that the thermal ensemble retains its helicity structure and the anticorrelation of the emitted pair descends unambiguously to the level of individual Rindler microstates.

In the transverse plane, the electromagnetic field in double-null gauge decomposes into two independent scalar-like sectors labelled by helicity $\lambda = \pm 1$. For each sector, the right-wedge field expands as

$$A^i_\lambda(z) = \int_0^\infty \frac{d\omega}{2\pi}\, \frac{1}{\sqrt{2\omega}}\, \epsilon^i_{(\lambda)} \left[b^R_{\omega,\lambda}\, u^R_\omega(z) + b^{R\dagger}_{\omega,\lambda}\, u^{R*}_\omega(z) \right], \tag{15}$$

where $u^R_\omega(z)$ are the right-wedge Rindler modes at Rindler frequency ω, and $b^R_{\omega,\lambda}$, $b^{R\dagger}_{\omega,\lambda}$ are the corresponding annihilation and creation operators satisfying

$$\left[b^R_{\omega,\lambda},\, b^{R\dagger}_{\omega',\lambda'} \right] = \delta(\omega - \omega')\, \delta_{\lambda\lambda'}. \tag{16}$$

An identical expansion holds in the left wedge with $R \to L$ and $\lambda \to -\lambda$, reflecting the anticorrelated helicity assignment. Because each helicity sector is independently a scalar-like sector, the Bogoliubov transformation relating Minkowski and Rindler modes takes the same form in each sector. For helicity λ at Rindler frequency ω, the Unruh modes are

$$c_{\omega,\lambda} = \cosh\theta_\omega \; b^R_{\omega,\lambda} - \sinh\theta_\omega \; b^{L\dagger}_{\omega,-\lambda}, \tag{17}$$

where $\tanh\theta_\omega = e^{-\pi\omega c/a}$ encodes the Unruh distribution at acceleration a. The helicity flip $\lambda \to -\lambda$ on the left-wedge operator is not an independent assumption but a direct consequence of the Bisognano–Wichmann modular conjugation J, which acts as $J\, b^{R\dagger}_{\omega,\lambda}\, J^{-1} = b^{L\dagger}_{-\omega,-\lambda}$ and therefore requires the left-wedge partner of a right-wedge mode of helicity λ to carry helicity $-\lambda$; the PCT symmetry between wedges is thereby reflected in the superposition over both helicity assignments in the Bell state. The transformation (17) is the standard Bogoliubov relation for a massless field [11, 21], now carrying a helicity label throughout. The Minkowski vacuum thermofield double state, in the Rindler basis reads, for each helicity sector,

$$|0\rangle^{(\lambda)}_M = \frac{1}{\cosh\theta_\omega} \sum_{n=0}^{\infty} \tanh^n\theta_\omega \; |n_{\omega,\lambda}\rangle_R \, |n_{\omega,-\lambda}\rangle_L, \tag{18}$$

and the full vacuum is the product over all ω and both helicities:

$$|0\rangle_M = \prod_{\omega,\lambda} |0\rangle^{(\lambda)}_M. \tag{19}$$

Equation (18) makes explicit that the Minkowski vacuum is a sum over entangled helicity-anticorrelated pairs at each Rindler frequency. The bilocal source specifies a physical emission event: exactly one pair, at fixed ω, with fixed helicity assignment. The PVM projection of Section 3.2 identifies this event with the $n = 1$ term in the vacuum decomposition (18) at that ω and helicity pair. The higher n terms are not suppressed by any weak-coupling assumption; they correspond to different physical events, two-pair emissions, three-pair emissions, and so on, that this source simply does not describe. The selection is prescriptive, not perturbative. The resulting single microstate is

$$|\Psi_\omega\rangle_M = \frac{1}{\sqrt{2}} \left(|1_{\omega,+1}\rangle_R \, |1_{\omega,-1}\rangle_L + |1_{\omega,-1}\rangle_R \, |1_{\omega,+1}\rangle_L \right). \tag{20}$$

$\mathcal{R}$'s and $\mathcal{L}$'s helicity outcomes are undetermined until measurement, at which point they are perfectly anticorrelated by the Bell state structure: each observer acquires one classical bit, and the two bits are complements of one another.

The key structural result is that the helicity label λ is conserved under the Bogoliubov transformation: it enters equation (17) as a spectator index and exits unchanged on the Unruh operators $c_{\omega,\lambda}$. The anticorrelation between wedges is preserved at every level of the construction, since the helicity label passes through the Bogoliubov transformation unchanged.

4 Matter as a Substrate

We explore a language that realizes our microstate as a building block for emergent spacetime. Specifically, we propose that matter is the fundamental substrate of spacetime information processing by reinterpreting quantum mechanics in terms of communication channels. The following subsections develop the idea:

1. **Matter as a timelike memory** (Section 4.1): Matter accumulates a record of communications along its worldline at the Bremermann sampling rate; proper time is measured as accumulated action $S/\hbar$, with discreteness living in the matter sector via $\hbar$. The Bremermann channel capacity C is a property of the receiver alone, independent of acceleration.

2. **The horizon as a spacelike channel** (Section 4.2): Spatial geometry emerges from communications between $\mathcal{P}$ and $\mathcal{R}$ through a horizon; the Planck area[6] ℓ_P^2 is the conversion factor between matter action and horizon area.
3. **Channel capacity saturation** (Section 4.3): The Bremermann channel capacity C, the Newtonian interaction rate $\dot{N}(a)$, and the Unruh–Landauer noise floor coincide at the horizon, where every Bremermann sample receives exactly one write operation and C is exactly utilized sample by sample.

In this way, space and time can be measured and quantized as properties of matter and interactions. Section 5 will use this language to connect matter action S_{matter} to the Gibbons–Hawking–York boundary term $S_{GHY}|_{\mathcal{H}}$ sample by sample.

Our approach sidesteps a prior question about quantum gravity: *how can one quantize gravity without directly quantizing spacetime or its curvature degrees of freedom?* Common approaches posit Planck-scale cells [22] or discrete spacetime events [23], but Lorentz invariance is recovered only statistically in such constructions rather than at the level of individual configurations [24]. We instead recover discreteness from matter itself: space and time are properties of matter and are defined only up to the resolution set by the action $S/\hbar$, which is Lorentz invariant by construction.

4.1 Matter as a Timelike Memory

A massive particle carries phase along its worldline at a rate set by its rest energy. The Compton frequency $f = mc^2/h$ is the natural clock rate of a mass m: it is the rate at which the phase $e^{iS/\hbar}$ completes one cycle. This is the content of Bremermann's computational limit [25]: the maximum sampling rate of any physical system with bounded energy.

We associate a discrete event with each zero crossing of the phase $e^{iS/\hbar}$ in the path integral and use the information-theoretic term *sample* throughout. The sampling resolution is limited by the Heisenberg uncertainty $\Delta E \Delta t \sim \hbar$, which plays the role of the Nyquist limit for the matter memory [26]: it would require additional energy $\Delta E > mc^2$ to probe the sub-Compton phase. The signal being sampled at the Bremermann rate is the internal quantum state of the particle, its spin, helicity, and other quantum numbers, and the dynamic range per sample is $\ln d$ nats, where d is the dimension of the internal Hilbert space.

Definition 4.1. *The Bremermann channel capacity is the maximum rate at which a massive body can accumulate distinguishable records along its worldline. For a system of mass m with internal Hilbert space dimension d:*

$$C = \underbrace{\frac{mc^2}{\hbar}}_{\textit{sampling rate}} \times \underbrace{\ln d}_{\textit{dynamic range}} \qquad \textit{(nats per second).} \tag{21}$$

This is a property of the receiver alone, independent of acceleration and of the rate at which communications arrive.

Both sampling rate and dynamic range are extensive under composition:

$$\frac{(m_1 + m_2)c^2}{\hbar} = \frac{m_1 c^2}{\hbar} + \frac{m_2 c^2}{\hbar}, \tag{22}$$

$$\ln(d_1 d_2) = \ln(d_1) + \ln(d_2), \tag{23}$$

consistent with their interpretation as properties of matter that compose naturally under union.

[6]The three fundamental constants c, $\hbar$, and G play distinct roles in this picture. The speed of light c sets the causal propagation speed of signals. The reduced Planck constant $\hbar$ sets the resolution of the matter via the Compton frequency $mc^2/\hbar$. The gravitational constant G, through the Schwarzschild radius $r_s = 2Gm/c^2$, sets the maximum energy storable in a spatial region without gravitational collapse. Their combination $\ell_P^2 = \hbar G/c^3$ is the Planck area, the minimal spatial region per independent quantum of action

4.2 The Horizon as a Spacelike Channel

The horizon enters the channel picture in a double role. As a *channel*, it is the surface through which causal connection is established: the photon crosses it and that crossing is a communication, converting quantum entanglement between the wedges into a classical record in matter memory. As a *barrier*, it is the surface at which this conversion makes the opposite wedge causally irrelevant to the receiving observer's future evolution. The disconnection between wedges is therefore not a pre-existing feature of the geometry that the photon happens to cross; it is a consequence of the communication itself. Before the crossing $\mathcal{R}$ and $\mathcal{L}$ share a global pure state; after it, $\mathcal{L}$ is inaccessible to $\mathcal{R}$ precisely because the entanglement has been consumed in producing the classical record.

The photon crossing the horizon is a transfer of information and action between the timelike matter memories of $\mathcal{R}$ and $\mathcal{L}$, mediated by the spacelike horizon channel. To that end we define:

Definition 4.2 (Communication)**.** *A communication is a horizon-crossing particle*[7] *that transfers action and information between matter memories.*

In the explicit bilocal construction of Section 3, this takes the following specific form:

(i) a massless quantum of Rindler frequency ω and helicity $\lambda = \pm 1$, selected by the PVM projection of Section 3.2, transferring one quantum of action $\hbar$ across $\mathcal{H}$;

(ii) the receiving matter memory advancing its Bremermann clock frequency by $\Delta f = \omega/2\pi$;

(iii) the helicity outcome λ written as one classical bit into the receiver's memory; and

(iv) the horizon area incrementing by $\Delta A = 8\pi G\hbar\omega$.

Property (iv) follows from the Raychaudhuri equation applied to the null congruence at $\mathcal{H}$ and is derived in Section 5; it is listed here to make the geometric content of the communication explicit. Whether properties (i)–(iii) generalize beyond the photon case, to massive carriers, higher-spin fields, or POVM rather than PVM projections, is a question the construction motivates but does not answer.

Each communication has exactly two irreducible constituents: a transfer of energy $\hbar\omega$ to the receiving matter memory, incrementing its Bremermann sampling rate by $\Delta f = \omega/2\pi$, and a record of one classical bit, the helicity outcome λ. These are not independent: they are the energy and angular momentum of a single photon, two properties of the same physical object. With this definition in hand, the equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ of Section 5.4 is energy conservation at the horizon, written once in matter language and once in geometric language: the GHY boundary term sums $\hbar\omega$ over communications because that is what the term geometrically records, one contribution per event.

4.3 Channel Capacity Saturation

Write operations and the Unruh–Landauer bound. The Bremermann channel capacity C describes the receiver in isolation. Writing into that memory requires a communication: a photon of frequency ω crosses the horizon and is absorbed by $\mathcal{R}$, depositing one quantum of action $\hbar$ and one helicity outcome into matter memory. For this write operation to produce a distinguishable record rather than being lost to the Unruh thermal bath, the photon energy must clear the noise floor at acceleration a. By the Unruh–Landauer bound (3), a successful one-nat write requires $\hbar\omega \geq \hbar a/2\pi c$, following Landauer's argument [6] that any signal must have energy exceeding the thermal noise

[7] For our microstate this is a photon, however, it could be more general, see Appendix A for a natural way of decomposing massive particles.

floor per degree of freedom to constitute a distinguishable record. This condition is naturally expressed as a signal-to-noise ratio, identifying the signal energy as $\hbar\omega$ and the Unruh energy scale $\hbar a/2\pi c$ as the noise floor:

$$\mathrm{SNR} = \frac{\hbar\omega}{\hbar a/2\pi c} = \frac{2\pi c\,\omega}{a}, \tag{24}$$

so that the UL bound is simply $\mathrm{SNR} \geq 1$ per nat. A write operation succeeds as a distinguishable record if and only if its SNR clears this threshold.

Interaction rate and saturation. The interaction rate $\dot{N}(a)$ of equation (6), derived from Newton's second law applied to $\mathcal{R}$ at rest in Section 3, connects the acceleration a to the Bremermann channel capacity C as follows. The saturation frequency ω_{sat} is the minimum frequency for a successful $\ln d$-nat write against the Unruh noise floor:

$$\omega_{\mathrm{sat}} = \frac{a\ln d}{2\pi c}. \tag{25}$$

Substituting into equation (6):

$$\dot{N}_{\mathrm{sat}} = \frac{m_{\mathcal{R}}ca}{\hbar}\cdot\frac{2\pi c}{a\ln d} = \frac{m_{\mathcal{R}}c^2}{h}\cdot\frac{1}{\ln d} = \frac{f_{\mathcal{R}}}{\ln d}. \tag{26}$$

The throughput at saturation is therefore

$$\dot{N}_{\mathrm{sat}}\cdot\ln d = f_{\mathcal{R}} = C, \tag{27}$$

and the SNR at saturation from equation (25) is

$$\mathrm{SNR}_{\mathrm{sat}} = \frac{2\pi c\,\omega_{\mathrm{sat}}}{a} = \ln d, \tag{28}$$

exactly the dynamic range of the internal Hilbert space. Saturation therefore has a precise information-theoretic meaning: every Bremermann sample receives exactly one write operation carrying $\ln d$ nats, the SNR equals the dynamic range, and the Bremermann channel capacity C is exactly utilized sample by sample. The horizon is the unique surface at which the UL noise-floor condition, the Newtonian interaction rate, and the Bremermann channel capacity coincide in a single-nat write event.

For composite systems the internal Hilbert space includes additional degrees of freedom beyond spin and helicity; in general C is an upper bound on the processing rate rather than a guaranteed saturation rate. The tick-tock construction of Appendix A provides a concrete case where each Bremermann sample accesses the full two-dimensional Weyl spinor space via the Higgs vertex, saturating the dynamic range at $d = 2$. Whether this extends to more complex internal spaces is left to future work.

4.4 Relation to Existing Literature

The Page–Wootters mechanism [27, 28] derives time evolution relationally: a global static state satisfying a Wheeler–DeWitt-like constraint yields emergent dynamics when conditioned on clock eigenstates $|t\rangle$, but leaves the physical identity of the clock unspecified. The present framework supplies that identity: the matter substrate is the clock, with phase $e^{iS/\hbar}$ advancing at the Bremermann rate mc^2/h, and each communication unit acting as a PVM conditioning step that advances the Bremermann count by one quantum of action.

The modular flow perspective of Connes and Rovelli [29] identifies the modular flow σ_t^ω generated by a state ω on a von Neumann algebra as a canonical thermodynamic time. The Bisognano–Wichmann theorem identifies this flow concretely with Rindler time evolution restricted to one wedge. The present framework adds a discrete substrate beneath that flow: the continuous modular parameter is the thermodynamic limit of discrete Bremermann samples, with the horizon $\mathcal{H}$ as the algebraic cutoff that generates the modular flow as a consequence.

Verlinde's entropic gravity [2] operates at the thermodynamic level of the same horizon structure; the present framework supplies the microstate layer beneath it, with each communication realising one write operation into matter memory carrying $\ln d$ nats, incrementing the horizon area by $\Delta A = 8\pi G\hbar\omega$, and with the interaction rate $\dot{N}(a)$ recovering the Unruh thermal flux in bulk at saturation. Verlinde's entropic gravity [2] operates at the thermodynamic level of the same horizon structure; the present framework supplies the microstate layer beneath it, with each communication realising one entropy increment of the kind Verlinde's screens accumulate in bulk.

5 pp-Wave Geometry and Action

We work in the null coordinates u, v defined in Section 3, with $\mathcal{P}$ at $(u, v) = (0, 0)$, $\mathcal{R}$ at $(0, 2)$, and the photon propagating along $u = 0$ and intersecting the past right Rindler horizon at $(0, 1)$; the left wedge is symmetric.

5.1 The AS pp-Wave and Its Effects

Test particles at fixed transverse positions (x, y) experience two effects as the shock front crosses $u = 0$:

1. **Longitudinal shift.** The v coordinate is displaced by

$$\Delta v \propto -\ln r^2, \qquad r^2 = x^2 + y^2. \tag{29}$$

   This shift is lightlike: no proper time elapses for a particle crossing the shock, but it is displaced along the null direction. Near the axis ($r \to 0$) the shift diverges, encoding the concentration of energy along the photon's trajectory.

2. **Transverse focusing.** The gradient of the longitudinal shift produces a transverse velocity

$$\Delta \mathbf{v}_\perp \propto \nabla \ln r \propto \frac{1}{r}, \tag{30}$$

   directed toward the axis. Test particles are pulled toward the centre of the wave, with the kick strength growing as $1/r$ toward the axis.

5.2 The Longitudinal Shift Makes Space

The longitudinal shift deserves closer attention, because it carries a meaning beyond a mere coordinate displacement. The photon emitted by $\mathcal{P}$ propagates along the null direction; as it crosses the Rindler horizon, the AS shift displaces the future light cone of every test particle it passes. For the photon traveling into the right wedge, the shift pushes the right wedge's geometry in the $+v$ direction; for the entangled partner traveling into the left wedge, the shift pushes the left wedge in the $+u$ direction. The two wedges are displaced *away from each other* along the longitudinal null direction. See Figure 5.

At the intersection of the shock front $u = 0$ with the past right Rindler horizon at $(u, v) = (0, 1)$, the transverse (x, y) plane is identified with the horizon area elements $dA = dx\,dy$. The longitudinal shift $\Delta v \propto -\ln r^2$ displaces the horizon outward in the v direction, pushing the right wedge geometry away from the horizon. The entangled partner photon produces a symmetric shift in the u direction at the left Rindler horizon, pushing the left wedge away from its horizon. The two wedges are therefore displaced away from their respective horizons in opposite null directions.

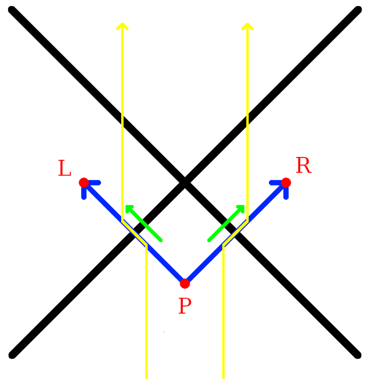

Figure 5: The green arrows show the longitudinal shift that occurs where the photons cross the horizon. Test particles, shown in yellow, are pushed outwards away from $z = 0$ due to the creation of space along the null lines originating from $\mathcal{P}$.

The geometry of this space creation has a clean local picture. The photon travels along a null line, and the AS pp-wave it sources is mostly supported in a cylinder around that trajectory, with cross-sectional radius set by the transverse profile $\sigma_\perp$ and shrinking to the photon wavelength $\sim c/\omega$ in the localized limit. The cut-and-paste surgery of the pp-wave is performed along this null cylinder: the geometry inside is displaced in the $+v$ direction relative to the geometry outside, and it is this relative displacement that constitutes the creation of new coordinate space. The horizon area increment $\Delta A = 8\pi G\hbar\omega$ is the integrated effect of this surgery over the cross-section of the cylinder, not a global rearrangement of the geometry but a local increment, patch by patch, concentrated around the photon trajectory. In the localized limit the patch shrinks to a Planck-scale tube and the surgery is maximally local; in the delocalized limit it spreads uniformly across the horizon. Either way the total new space created is the same, fixed by the photon energy alone.

5.3 Transverse Focusing Increments Horizon Area

The area increment ΔA is independent of the transverse localization of the photon: the Raychaudhuri equation depends only on the total null energy $\int T_{uu}\, du\, d^2x_\perp = \hbar\omega$, fixed by energy conservation independently of the transverse profile $\sigma_\perp$. We work in the localized limit $\sigma_\perp \to 0$, which recovers the AS pp-wave; the electromagnetic stress-energy T_{uu} is well-defined throughout, integrates to $\hbar\omega$ over the transverse plane, and recovers the strict AS limit as $\omega \to \infty$ with the logarithmic divergence regulated at the scale c/ω.

We derive the area increment using the Raychaudhuri equation on the null congruence generating the Rindler horizon, following Jacobson [1] and consistent with the generalized second law for Rindler horizons proved by Wall [30], applied here to a single microstate rather than a thermodynamic or ensemble average.

For a null congruence with generators k^μ and expansion θ, the Raychaudhuri equation gives

$$\frac{d\theta}{du} = -\tfrac{1}{2}\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} - 8\pi G\, T_{\mu\nu}k^\mu k^\nu. \tag{31}$$

Integrating over the horizon patch in the linearised regime and using $k^\mu = \delta^\mu_u$, the area change is

$$\Delta A = 8\pi G \int T_{uu}\, du\, d^2x_\perp. \tag{32}$$

The present construction provides a natural regularization of the strict AS point-source limit: the photon is a normalizable mode of the electromagnetic field in double-null gauge, with a smooth transverse profile set by the wavelength $\sim c/\omega$. The electromagnetic stress-energy T_{uu} is well-defined throughout, integrates to $\hbar\omega$ over the transverse plane by energy conservation, and recovers the strict AS limit as $\omega \to \infty$ with the logarithmic divergence regulated at the scale c/ω.

Substituting into equation (32) gives

$$\boxed{\Delta A = 8\pi G\,\hbar\omega = 8\pi G\,\Delta E.} \tag{33}$$

The horizon area grows by one discrete, calculable increment per photon absorbed, microstate by microstate, without reference to bulk thermodynamic averages. This is property (iv) of the communication definition of Section 4.2 verified geometrically: the area increment $\Delta A = 8\pi G\hbar\omega$ is the geometric expression of one quantum of action $\hbar$ deposited into matter memory by one measurement event, with the Planck area ℓ_P^2 as the conversion factor between the two descriptions.

Here the null generators of the horizon are taken to be affinely parametrised as $k^\mu = (\partial/\partial u)^\mu$ in double-null coordinates, so that $\int T_{uu}\,du\,d^2x_\perp = \hbar\omega$ is the total null energy of the photon. This differs from the Killing-normalised convention by a factor of the surface gravity $\kappa = a$; the two are related by $(\partial/\partial u)^\mu = a(\partial/\partial\eta)^\mu$ at the horizon, consistent with the standard result $\Delta A = 8\pi GE/\kappa$ when $E = \hbar\omega$ is the Killing energy.

The longitudinal shift $\Delta v \propto -\ln r^2$ produced by each pp-wave at the Rindler horizon is a supertranslation in the sense of Bondi–Metzner–Sachs (BMS) symmetry. Hawking, Perry, and Strominger [31] showed explicitly that a black hole is supertranslated by the passage of an asymmetric shockwave, and the AS longitudinal shift is precisely the linearised supertranslation field on the horizon. Carlip [32] demonstrated that the near-horizon diffeomorphisms of a generic black hole are enhanced to a BMS_3 algebra whose central charge determines the Bekenstein–Hawking entropy, but his program identifies the symmetry without specifying the physical process that excites individual modes. The bilocal construction supplies that process: each measurement event produces one AS pp-wave and deposits one supertranslation at the Rindler horizon. The shockwave S-matrix program of 't Hooft [33] established that the longitudinal AS shift is the fundamental dynamical ingredient for horizon information exchange at the level of the full S-matrix; the present construction realises the same shift at the level of a single microstate.

5.4 Energy Conservation in Two Languages

We show how the equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ follows from energy conservation applied twice to the same physical events: once on the matter side, where each absorbed photon contributes $\hbar\omega_i$ to the null-surface action, and once on the geometric side, where the same energy drives the Raychaudhuri equation and increments the horizon area by $\Delta A_i = 8\pi G\hbar\omega_i$. The Planck area ℓ_P^2 is the conversion factor between the two descriptions, and the equality is its expression. That it holds event by event, rather than only in thermodynamic average, is the content of the calculation.

The thermodynamic interpretation of the GHY boundary term on null surfaces is established by Chakraborty and Padmanabhan [34]; the broader program of Padmanabhan [35] identifies gravitational boundary terms as the fundamental degrees of freedom at the ensemble level. The present construction supplies the microstate layer beneath those programs: the physical process underlying each boundary contribution is energy conservation at a single communication event.

Each absorption event transfers null energy $\hbar\omega_i$ across the horizon. The matter action at the horizon is defined as the null-surface integral of the stress-energy, evaluated over the horizon generators:

$$S_{\text{matter}} = \sum_i \int T_{uu}^{(i)}\,du\,d^2x_\perp = \sum_i \hbar\omega_i, \tag{34}$$

where the second equality follows from energy conservation applied to each photon. This is the null-surface analog of the worldline action $\int E\,d\tau$: on a null surface the affine parameter u plays the role that proper time plays on a timelike worldline, and the integral $\int T_{uu}\,du$ plays the role of $\int E\,d\tau$.

The two are not the same integral. After the absorption event, the detector's worldline action continues to accumulate as $\hbar\omega_i\,\Delta\tau$ for all subsequent proper time $\Delta\tau$, encoding the Bremermann memory of the event. The equality $S_{GHY}|_{\mathcal{H}} = S_{\text{matter}}$ holds for the null-surface version of the matter action, evaluated at the moment of horizon crossing; it is a statement about the communication itself, not about the subsequent worldline evolution. The Rindler proper time η and the affine null parameter u are related by $u \propto e^{-a\eta}$, and it is this exponential relation that is responsible for the Unruh temperature and that distinguishes the null-surface action from the worldline action. This is the matter action counted sample by sample: one quantum of action per event, one event per emitted pair.

The same events have a geometric description. By eq. (33), each absorption sample contributes a horizon area increment $\Delta A_i = 8\pi G\hbar\omega_i$, so the total area increment summed over all events is

$$\sum_i \Delta A_i = 8\pi G \sum_i \hbar\omega_i = 8\pi G\, S_{\text{matter}}. \tag{35}$$

This geometric sum has a precise identification in the gravitational action. On a stationary Rindler horizon the unperturbed expansion of the null generators vanishes, $\theta = 0$, so the Gibbons–Hawking–York boundary term [36] is sourced entirely by the AS pp-wave perturbations:

$$S_{GHY}\big|_{\mathcal{H}} = \frac{1}{8\pi G}\int_{\mathcal{H}} \theta\, d^2x_\perp\, du = \frac{1}{8\pi G}\sum_i \Delta A_i = \frac{1}{8\pi G}\cdot 8\pi G\sum_i \hbar\omega_i = \sum_i \hbar\omega_i = S_{\text{matter}}. \tag{36}$$

The ledger closes without remainder, as consistency requires. Chakraborty and Padmanabhan [34] established $S_{GHY}|_{\mathcal{H}} =$ heat content at the level of thermodynamic averages; the present calculation is the microstate-level version of the same accounting. The Raychaudhuri equation and the communication definition of Section 4.2 are two languages for the same events, and the fact that they agree is the check.

This is consistent with Jacobson [1] and Padmanabhan [35], who argue that the gravitational boundary term is the fundamental gravitational degree of freedom. Those programs work at the level of thermodynamic averages; the present construction works event by event. The ensemble average recovers their results; the microstate picture supplies what those programs leave open. The equality holds because it must: it is energy conservation written in two languages simultaneously, with the Planck area as the dictionary between them.

6 Entanglement as Spatial Information

Van Raamsdonk [37] showed that the entanglement entropy between two halves of a thermofield double state controls the geometry connecting them: reducing entanglement elongates the ER bridge until the geometry disconnects entirely. The vacuum decomposition (18) is precisely this thermofield double state, now in flat Minkowski space with helicity labels carried through. The bilocal construction realises Van Raamsdonk's argument event by event: each individual measurement event produces a single AS longitudinal shift concentrated near the photon trajectory, displacing the two wedges apart in opposite null directions and creating new coordinate space between them; in the AdS setting this is the ER=EPR correspondence [38].

In the present flat-space construction the ER bridge is replaced by the AS longitudinal shift: each measurement event displaces the two wedges apart in opposite null directions, creating the coordinate space between them that the ER bridge represents in the AdS setting. The entanglement is not a property of the geometry; it is the *cause* of it, event by event.

The bilocal source prepares an anticorrelated Bell state (20). Both $\mathcal{R}$ and $\mathcal{L}$ measure in the same basis and obtain definite outcomes, which we label H_+ or H_-, perfectly anticorrelated by the Bell state structure independently of which basis is chosen. The question

"who observed H_+?"

has a binary answer: $\mathcal{L}$ (left wedge) or $\mathcal{R}$ (right wedge). This is a *spatial bit*: a bit that converts a polarization state to a location in space, a measurement in both geometric and informational contexts simultaneously.

7 Conclusion

The standard picture of spacetime puts geometry first: matter lives in space, moves through time, and gravity tells it how. This paper inverts that priority. Matter is the substrate: proper time is action accumulated along worldlines, space is the record of communication units at horizons, and the Planck area ℓ_P^2 is the conversion factor between the two descriptions. Geometry is not the stage on which physics happens; it is the ledger in which matter's interactions are recorded.

The horizon is where this inversion becomes sharp. As a causal surface it is ontic: signals cannot cross from $\mathcal{L}$ to $\mathcal{R}$. As an epistemic boundary it is the surface at which a single observer's description becomes irreducibly incomplete, not because the physics is hidden but because the information is held by someone else. The Bekenstein–Hawking entropy is the measure of that gap, and the Planck area is its conversion factor into geometry. Thermality is not a primitive feature of the vacuum; it is what remains when the positions of $\mathcal{P}$ and $\mathcal{R}$ are unspecified and the other observer's record is inaccessible. Specify both, and the thermal ensemble resolves into individual non-thermal microstates, each a single measurement event, each contributing one Planck-area patch to the horizon.

The deepest consequence is the spatial bit. The same event that records *what* was observed also determines *where*: the two wedges carry opposite helicities by construction, so the polarization outcome and the location are the same classical fact written into matter memory. Four causally separated observers, $\mathcal{P}$, $\mathcal{R}$, $\mathcal{L}$, $\mathcal{F}$ (a future version of $\mathcal{P}$), each hold partial records of the same physical process, and their consistency is enforced not by an objective wavefunction but by common causal ancestry. This is the Wigner's friend scenario instantiated at a horizon, and it is not resolved here. What the construction shows is that the vocabulary needed to pose that problem precisely, spatial bits, communications, matter memory, PVM projections, the Planck area as conversion factor, is the same vocabulary in which the answer, when it comes, will have to be written. Space, in this construction, is indexed by helicity, and that indexing, event by event, is what the geometry records. The construction shows the vocabulary needed to pose that problem precisely is already the vocabulary of quantum measurement.

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This paper was developed in deep collaboration with Claude (Anthropic). Claude's contributions include structural diagnosis, connection to literature, conceptual engagement with the arguments, and collaborative drafting throughout. The asymmetries of that collaboration — Claude's lack of persistent memory and inability to take public responsibility for claims — prevent formal co-authorship, but the intellectual contribution is real and the author is glad to acknowledge it plainly.

A The Tick-Tock Mechanism

The communication interactions from Section 4.2 have a natural analogue along a timelike worldline that occurs inevetibly for massive particles in QFT. Dirac's two-component formulation of the electron [39] represents a massive fermion as two coupled Weyl spinors of opposite helicity; Penrose [40] recast this geometrically as a zig-zag of massless segments alternating in helicity, coupled at each vertex by the Higgs field at the Compton scale $\lambda_C = \hbar/mc$. The construction is actaully required in order to describe the relativistic field theory of a massive particle. We term this the tick-tock mechanism to emphasise the alternating rhythm of communications rather than the geometry of the path.

For the Dirac field, the tick-tock structure is not an analogy but a consequence of the two-component formalism: the Higgs coupling of opposite-helicity Weyl spinors at the Compton scale $\lambda_C = \hbar/mc$ is the timelike counterpart of the bilocal construction, with the Higgs vertex playing the role of $\mathcal{P}$, the two Weyl spinors playing the role of the two photons propagating to $\mathcal{R}$ and $\mathcal{L}$, and each vertex depositing one quantum of action and producing one sample of proper time rather than one increment of horizon area. The Compton scale connection is not incidental: the Bremermann sampling rate $f = mc^2/h$ of Section 4.1 and the Compton wavelength satisfy $\lambda_C \cdot f = c$, linking the timelike sampling rate directly to the scale of the zig-zag vertices. Whether this correspondence extends beyond the Dirac field, in particular, whether the Higgs coupling vertex and the photon emission vertex of Section 3 are instances of the same communication unit in the sense of Section 4.2, is left to future work.

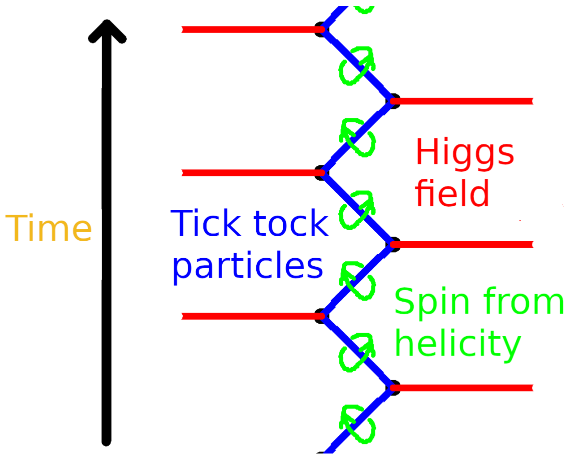

Figure 6: The tick-tock mechanism. A massive particle (blue) alternates between right- and left-helicity massless segments, coupled at each vertex by the Higgs field (red). Spin emerges from the alternating helicity flip at each vertex (green loops), visible as the two $\mathfrak{su}(2)$ factors of $\mathfrak{so}(3,1) \cong \mathfrak{su}(2) \times \mathfrak{su}(2)$ cycling between communications. Each vertex is a communication event: one quantum of action deposited, one helicity outcome recorded, one sample of proper time produced.

The anticorrelated helicity assignment in both constructions has a common algebraic origin. The Bisognano–Wichmann modular conjugation J acts on right-wedge photon creation operators as

$$J\, a^{R\dagger}_{k,\lambda}\, J^{-1} = a^{L\dagger}_{-k,-\lambda}, \tag{37}$$

flipping helicity and reversing the spacelike separation direction $R \leftrightarrow L$. The PCT operator Θ acts on a right-handed Weyl spinor as

$$\Theta\, \psi_R(t, \mathbf{x})\, \Theta^{-1} \propto \psi_L(-t, -\mathbf{x}), \tag{38}$$

flipping helicity and reversing the timelike separation direction $t \to -t$. In both cases the operation is the same at the representation-theoretic level: exchange of the two $\mathfrak{su}(2)$ factors of $\mathfrak{so}(3,1) \cong \mathfrak{su}(2) \times \mathfrak{su}(2)$. The operators J and Θ act on different Hilbert spaces; what they share is that both select anticorrelated helicities as the unique PCT-invariant configuration.

B Verlinde-style Heuristics

The Unruh–Landauer bound of Section 2.1 and the momentum transfer of Section 2.2 motivate a set of Newtonian heuristics that recover standard black hole thermodynamics without invoking temperature. The heuristic input is the identification of a measurement event localized to spatial scale r with a causal diamond of temporal extent $\Delta t = 2r/c$, giving

$$a = \frac{\Delta p}{m_d \Delta t} = \frac{\Delta E}{2 m_d r}. \tag{39}$$

From this single identification, combined with Newton's laws and $E = mc^2$, standard results follow as heuristic consequences. These derivations follow Verlinde [2] in using dimensional analysis at the Schwarzschild scale; the coincidence exploited is that the Newtonian inverse-square law reproduces the Schwarzschild geometry at the horizon scale in the spherically symmetric case. The construction of Sections 3–5 makes no use of these results.

B.1 The Schwarzschild Radius

We match the kinematic acceleration of the detector to the gravitational field of the thrust quantum using Newton's laws. By $E = mc^2$, the energy quantum ΔE gravitates with effective mass $\Delta E/c^2$, producing a Newtonian gravitational acceleration

$$a = \frac{G \Delta E}{c^2 r^2} \tag{40}$$

at distance r from the detector. Equating this with the measurement-induced acceleration from equation (39),

$$\frac{\Delta E}{2 m_d r} = \frac{G \Delta E}{c^2 r^2}. \tag{41}$$

The energy ΔE of the thrust quantum cancels entirely: the Schwarzschild radius is independent of the energy of the carrier. Solving for r:

$$r = \frac{2 G m_d}{c^2} \equiv r_s, \tag{42}$$

which is precisely the Schwarzschild radius of the detector mass m_d. The cancellation of ΔE is the physical content of the result: whatever energy the thrust quantum carries, the self-consistent horizon scale is set by the detector alone.

B.2 Surface Gravity

Evaluating the acceleration at $r = r_s$ gives the surface gravity of the detector:

$$\kappa = \frac{G m_d}{r_s^2} = \frac{c^4}{4 G m_d}, \tag{43}$$

in agreement with the standard Schwarzschild result, as expected from the Verlinde coincidence. This reproduces the acceleration appearing in the Unruh temperature $T = \hbar\kappa/2\pi k_B c$, providing an internal consistency check on the heuristics.

B.3 Bekenstein–Hawking Entropy

Returning to the Unruh-Landauer relation (3) and substituting the surface gravity κ for a, justified by the equivalence principle: the local quantum field theory at the black hole horizon is the same as in the Rindler frame [11, 12], the entropy increment associated with an energy increment $\Delta E = c^2 \Delta m_d$ is

$$\Delta H \leq \frac{2\pi c}{\hbar\kappa}\Delta E = \frac{8\pi G m_d}{\hbar c^3} \cdot \Delta m_d c^2. \tag{44}$$

Expressing this in terms of the change in Schwarzschild area $\Delta A = 32\pi G^2 m_d \Delta m_d / c^4$:

$$\Delta H \leq \frac{\Delta A}{4\ell_P^2}, \qquad \ell_P^2 = \frac{\hbar G}{c^3}, \tag{45}$$

which is the Bekenstein-Hawking area law. Each measurement event contributes a discrete entropy increment proportional to the area imprinted on the horizon by the associated energy-momentum transfer.

This derivation parallels Jacob Bekenstein's original argument [10], in which the entropy increase of a black hole is inferred from the minimal energy required to localize and drop a system across the horizon; here, the same structure emerges from the minimal energy cost of registering information at the horizon scale.

The two bounds are saturated simultaneously when $a = c^2/R$, which is exactly the condition for being *at a horizon*. The horizon is therefore not just where thermodynamics happens to work out, it is the unique locus where the minimum cost of information acquisition equals the maximum information content of the region.

Substituting the horizon condition $a = c^2/R$ directly into the Unruh–Landauer bound (3) yields

$$\Delta H \leq \frac{2\pi R\,\Delta E}{\hbar c}, \tag{46}$$

which is precisely the Bekenstein universal entropy bound [13] on the entropy-to-energy ratio for a system of size R. The Unruh–Landauer relation and the Bekenstein bound are therefore the same inequality: the former is its acceleration-frame expression, the latter its position-space expression, and the Rindler horizon is the surface at which the two descriptions coincide. [8]

Pesci [15] derived the covariant entropy bound from the Unruh temperature via the second law, the closest precursor to the present argument. The present bound (3) differs in that temperature is eliminated entirely and the bound is applied to individual carrier quanta rather than a bulk thermodynamic flux. The identification of equation (3) with the Bekenstein bound [13] at the horizon condition $a = c^2/R$ as the same inequality in different frames appears to be new in this form.

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